Isosceles and equilateral triangles

1

a Yes b No c Yes d Yes

2

a $x = 15$ b $x = 4$

3

a $x = 58$ b $x = 54$ c $x = 60$ d $x = 4$

e $x = 94$

4 $x = 71^{\circ}$

5

a
i $a = 45$ ii $b = 9$

b
i $a = 76$ ii $b = 5$

c
i $a = 59$ ii $b = 62$

d
i $a = 60$ ii $b = 18$

e
i $a = 62$ ii $b = 28$

6 $y = 30$

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To prove: $\angle HEF \cong \angle HGF$

	Statements	Reasons
1.	$\overline{FG} \cong \overline{EF}$	Given
2.	$\overline{FH}$ bisects $\angle EFG$	Given
3.	$\angle GFH \cong \angle EFH$	Definition of angle bisector
4.	$\overline{FH} \cong \overline{FH}$	Reflexive property
5.	$\triangle GFH \cong \triangle EFH$	SAS theorem
6.	$\angle HEF \cong \angle HGF$	Corresponding angles of congruent triangles are congruent

Additional questions

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a An isosceles triangle has two base angles which are congruent. The two sides opposite the base angles are also congruent and are called the legs of the triangle.

b An equilateral triangle has three congruent angles and three congruent sides.

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a True b False c True d True

e False f True

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a
- i $x = 70$
- ii The base angles are equal by the "base angles theorem"

b
- i $x = 60$
- ii The triangle is equiangular by the "corollary to the base angles theorem"

c
- i $x = 15$
- ii The two are congruent legs of an isoscles triangle by the "converse of base angles theorem"

d
- i $x = 8$
- ii The triangle is equilateral by the "corollary to the converse of base angles theorem", noting that the triangle is equiangular

e
- i $x = 128$
- ii Using the "triangle sum theorem" after noting that the triangle has two congruent base angles by the "base angles theorem"

f
- i $x = 62$
- ii The triangle has two congruent base angles by the "base angles theorem", then using the "triangle sum theorem" to calculate the angle measure

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a
- i $x = 70$ ii $y = 40$

b
- i $x = 8$ ii $y = 5$

12 $x = 4$

13 $y = 34$

15 The congruent pairs are:

- (i) and (vii)
- (ii) and (viii)
- (iii) and (vi)

16 Farah's stride length is 27 inches.

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a True. All equilateral triangles have three angles with a measure of 60°. So all the angles of an equilateral triangle are acute, therefore all equilateral triangles are acute.

b False. For example, an isosceles triangle could have angles with the measures 20°, 20° and 140°. Not all the angles in the triangle are acute, therefore not all isosceles triangles are acute. Corrected statement: Not all isosceles triangles are acute.

c False. A triangle with the angle measures 59°, 60° and 61° is acute but not equilateral. Corrected statement: If a triangle is equilateral, then it is acute.

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To prove: $\angle ABC \cong \angle ACB$

	Statements	Reasons
1.	$\overline{AB} \cong \overline{AC}$	Given
2.	$AB = AC$	Definition of congruent segments
3.	$m\angle BAD = m\angle CAD$	AD is the angle bisector of $\angle BAC$
4.	AD is common in $\triangle ABD$ and $\triangle ACD$	Given
5.	$\triangle ABD \cong \triangle ACD$	Side-Angle-Side triangle congruence test
6.	$\angle ABC \cong \angle ACB$	Corresponding angles in congruent triangles are congruent

This proves the base angles theorem: that if two sides of a triangle are congruent, then the angles opposite them are congruent.

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To prov[redacted] $\cong \overline{AC}$

	Statements	Reasons
1.	$\angle ABC \cong \angle ACB$	Given
2.	$m\angle ABC = m\angle ACB$	Definition of congruent angles
3.	$m\angle BAD = m\angle CAD$	AD is the angle bisector of $\angle BAC$
4.	AD is common in $\triangle ABD$ and $\triangle ACD$	Given
5.	$\triangle ABD \cong \triangle ACD$	Angle-Angle-Side triangle congruence test
6.	$\overline{AB} \cong \overline{AC}$	Corresponding sides in congruent triangles are congruent

This proves the converse of base angles theorem: that if two angles of a triangle are congruent, then the sides opposite them are congruent.

20 Since an equilateral triangle has all sides equal, we can apply the base angles theorem to any pair of sides. This can be used to show that an equilateral triangle is equiangular. Similarly, since an equiangular triangle has all angles congruent, we can apply the converse of base angles theorem to any pair of angles. This can be used to show that an equiangular triangle is equilateral.